

Aryaman Sharma

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EDUCATION

University of Waterloo

B.C.S Bachelor of Computer Science Honours Co-op (GPA: 3.70/4)

Waterloo, ON

Expected April 2030

- Awards: President's Scholarship of Distinction | 1x Term Distinction
- Coursework: Data Structures & Algorithms, Designing Functional Programs, Tools for Software Development

EXPERIENCE

Tutor - Mathematics & Computer Science

September 2023 - June 2025

Fletcher's Meadow Secondary School

Brampton, ON

- Delivered **150+** hours of support in computer science, teaching core **object oriented programming** concepts in **Python/Java** and guiding peers through coding projects, which boosted coding proficiency and problem solving.
- Tutored **15+** peers in **calculus** by simplifying complex concepts like **limits, derivatives, and integrals** into accessible explanations, reinforcing students' strategies while leading to **20-30% improvements** in test scores.

PROJECTS

Heart Disease Prediction | *Python, scikit-learn, NumPy, Pandas, Git*

[GitHub](#)

- Developed an end-to-end machine learning pipeline to predict heart disease risk. achieving **85% classification accuracy**, using logistic regression on **300+ patient records**.
- Improved model reliability through data cleaning and feature engineering, validated via an **80/20 train-test split**, using Python, Pandas, NumPy, and scikit-learn.
- Evaluated model performance on a held-out test using accuracy and **classification metrics**, ensuring robust generalization, with scikit-learn's evaluation framework.

Portfolio Website | *React, JavaScript, HTML/CSS, GitHub Pages*

[GitHub](#)

- Designed and developed a fully responsive personal portfolio using **React and modern JavaScript**, showcasing projects, about me, and contact details in an organized UI.
- Improved performance and accessibility through reusable components, semantic HTML, optimized assets, and mobile-first CSS layouts, ensuring smooth **cross-device rendering**.
- **Deployed** and maintained the site via GitHub Pages with **Git-based version control**, enabling seamless updates and continuous improvement.

RC Vehicle | *C++, Arduino, Embedded Systems, Electronics*

- Designed and built an RC car using Arduino microprocessor, C++ firmware using structured, modular code and interrupt-based logic, achieving a top speed of **15 km/h** with low-latency real-time motor control.
- Integrated embedded systems components for wireless operation, delivering a **30m control range** with stable signal performance and reliable system responsiveness.

Blackjack | *Python, Pygame, Git*

[GitHub](#)

- Built a fully interactive Blackjack game with real-time rendering and input handling, supporting **10+ game states** and complete win/loss/draw resolution, using Python and Pygame.
- Implemented robust game logic with hit/stand actions, automated dealer behavior (hit until 17), score calculation, and bust detection, scaling gameplay to **200+ cards** with randomized shuffling, using object oriented design.
- Designed a user-friendly graphical interface with restart/replay functionality, ensuring smooth gameplay and maintainability, while managing iterative development and bug fixes via Git with structured commits.

TECHNICAL SKILLS

Languages: Python, C++, C, Java, JavaScript, SQL (MySQL), HTML, CSS, Racket

Technologies: Git, Pandas, Scikit-learn, NumPy, Pygame, Matplotlib, React, NodeJS

Developer Tools: GitHub, VS Code, IntelliJ IDEA, PyCharm, Eclipse, Linux, Vim, Power BI, Microsoft 365

ACHIEVEMENTS

Certificate of Distinction in the University of Waterloo's Sir Issac Newton Exam (Physics)

May 2025

Certificate of Distinction in the University of Waterloo's Avogadro Exam (Chemistry)

May 2024